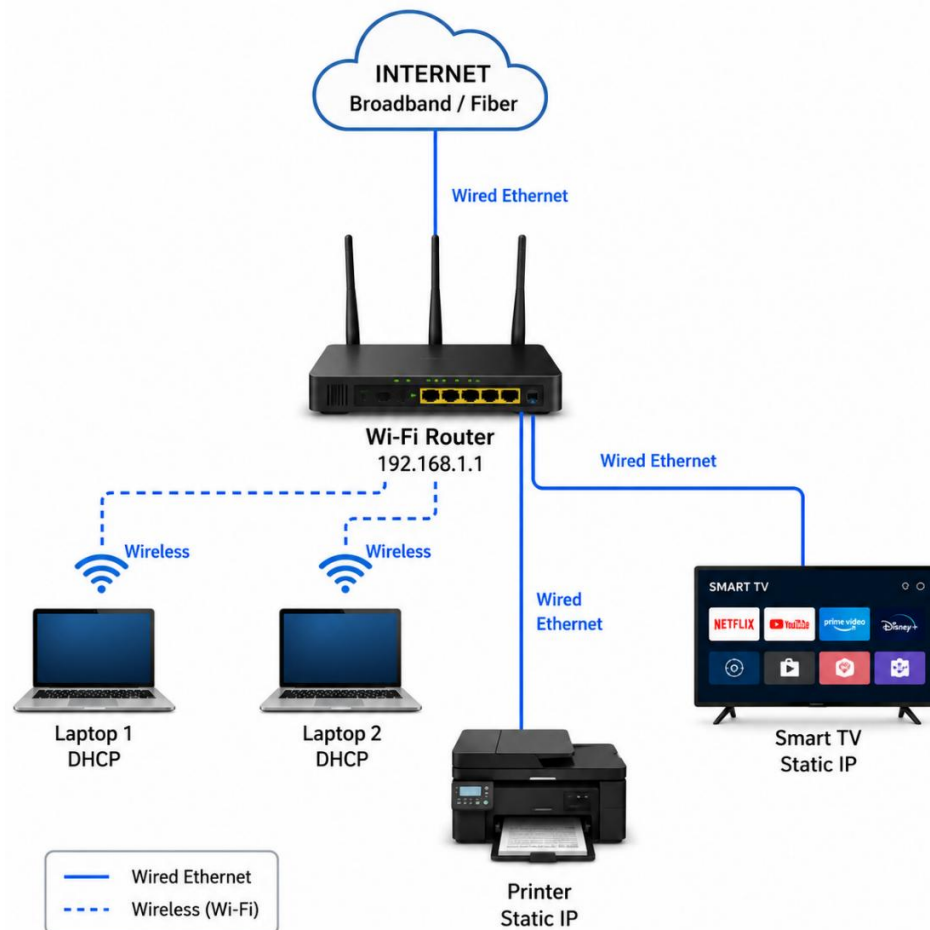


1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed
- Hint:** Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices

Component	Suggested Brand/Model	Qty	Purpose
Broadband/Fiber connection	Airtel/Jio/ACT or local ISP	1	Internet access
Wi-Fi Router	TP-Link Archer C6 AC1200	1	Wi-Fi + wired LAN
Laptop	HP/Dell/Lenovo	2	User devices
Smart TV	Samsung/LG/Sony	1	Network streaming
Network Printer	HP/Canon/Epson	1	Shared printing
RJ45 connectors	D-Link/Digisol	As needed	Ethernet termination

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

- Connect the ISP broadband/fiber modem or ONT to the router's **WAN/Internet port** using an Ethernet cable.
- Connect a laptop to the router through Wi-Fi or a LAN cable.
- Open the router's management page. Depending on the router, this may be an address such as 192.168.0.1 or 192.168.1.1.
- Log in using the router's initial administrator credentials.
- Open **Wireless/Wi-Fi Settings**.
- Change the SSID to a unique name, for example: **SOHO_Office_WiFi**.

- Open **Wireless Security** and select **WPA2-Personal/WPA2-PSK** (or WPA2/WPA3 if supported).
- Create a strong Wi-Fi password, rather than keeping the factory password.
- Go to the router's **Administration/System** settings and change the default router admin password.
- Enable the **DHCP Server** so laptops can receive addresses automatically.
- Save/apply the configuration and reconnect devices to the new SSID.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	Connection	MAC Address	IP Assignment	IP Address
Wi-Fi Router	WAN/LAN	A4:21:8C:11:20:01	Static	192.168.1.1
Laptop 1	Wireless	3C:52:82:AA:10:21	DHCP	192.168.1.101
Laptop 2	Wireless	8A:34:BC:20:55:12	DHCP	192.168.1.102
Smart TV	Wired	B8:27:EB:44:31:90	Static	192.168.1.20
printer	Wired	00:1A:2B:3C:4D:5E	Static	192.168.1.10

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Possible Cause	Diagnosis	Solution
Wi-Fi disconnected/wrong password	Check Wi-Fi icon and SSID	Connect to SOHO_Office_WiFi using the correct password
Incorrect IP configuration	Run ipconfig and check the IP	Set IPv4 to obtain IP/DNS automatically, then reconnect

Weak Wi-Fi signal	Move laptop near router and test	Reposition router or install Wi-Fi extender
DHCP problem	Check whether laptop receives a valid 192.168.1.x address	Restart adapter/router and renew DHCP lease
DNS problem	Internet IPs work but websites do not load	Check automatic DNS or configure a trusted DNS resolver
Laptop Wi-Fi adapter issue	Check Device Manager/network adapter	Disable/enable adapter or update/reinstall driver